USN [ ] [22AS41]

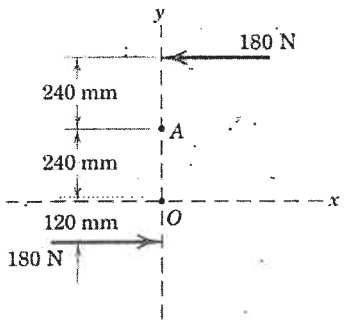
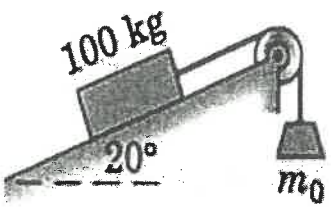
FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

ENGINEERING MECHANICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**05 x2=10**
1.	**a.**	Discuss the properties of the vectors	L2	CO1	**02**
	b.	Differentiate between coefficient of static and dynamic friction	L4	CO2	**02**
	c.	Define the following: area moment of inertia, product of inertia and mass moment of inertia.	L1	CO3	**02**
	d.	List the equations of motions	L1	CO4	**02**
	e.	State principle of virtual work	L1	CO5	**02**
		PART-B			**06 X 5=30**
2.	**a.**	A force is specified by the vector **F** = 50**i** - 40**j** + 60**k**lb. Calculate the angles made by **F** with the -x, -y, -z axes.	L3	CO1	05
		OR			
	b.	Compute the combined moment of the two I80-N forces about (a) point O and (b) point A 	L3	CO1	**05**
3.	**a.**	Determine the range of values which the mass $\mathbf{m_o}$ may have so that the 100-kg block shown in the figure will neither start moving up the plane nor slip down the plane. The coefficient of static friction for the contact surfaces is 0.30. 	L3	CO2	**05**
		OR			

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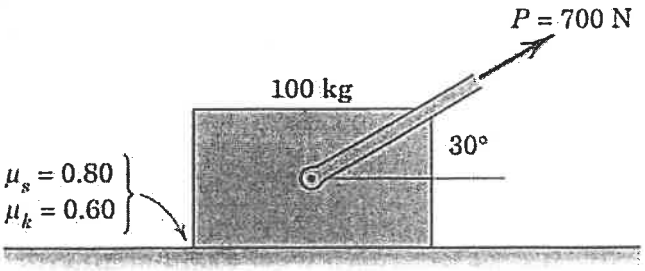
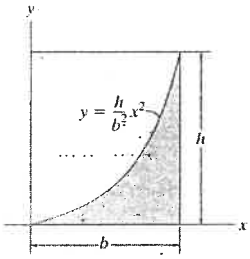
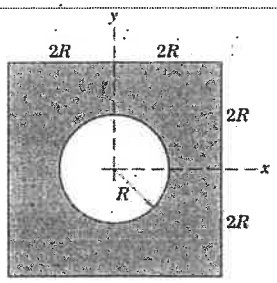
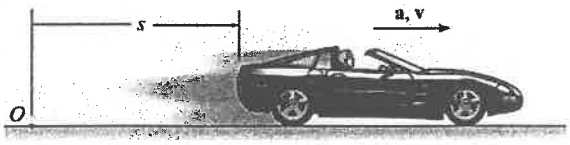
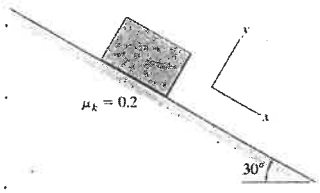
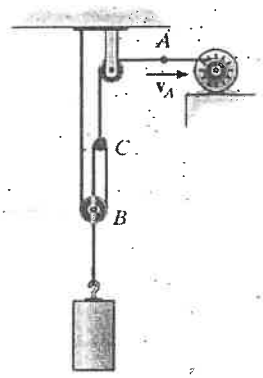
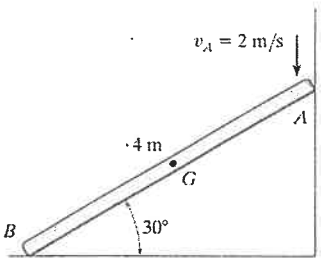
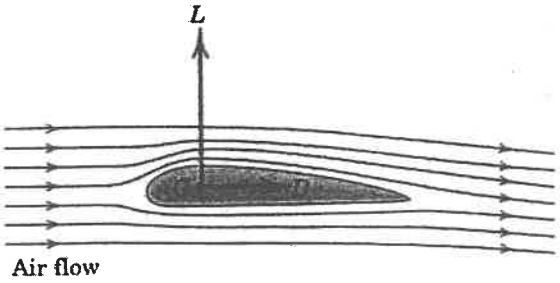
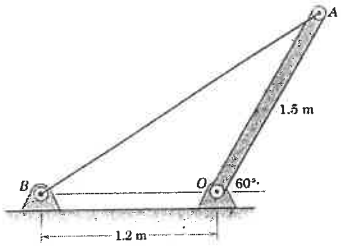
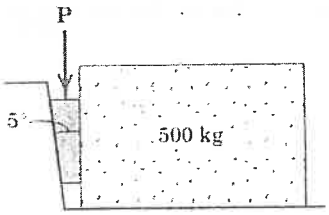
	b.	The 700-N force is applied to the 100-kg block, which is stationary before the force is applied. Determine the magnitude and direction of the friction force **F** exerted by the horizontal surface on the block. 	L4	CO2	**05**
4.	**a.**	Determine the location the centroid of the shaded area. 	L3	CO3	**05**
		OR			
	b.	Determine the moment of inertia about the *x*-axis of the square area without and with the central circular hole. 	L3	CO3	**05**
5.	**a.**	The car in figure moves in a straight line such that for a short time its velocity is defined by v = $(3t^2 + 2t)$ ft/s, where t is in seconds. Determine its position and acceleration when t = 3s. When t = 0, s = 0. 	L3	CO4	**05**
		OR			
	b.	During a test a rocket travels upward at 75 m/s, and when it is 40m from the ground its engine fails. Determine the maximum height S_B reached by the rocket and its speed just before it hits the ground. While in motion the rocket is	L4	CO4	**05**

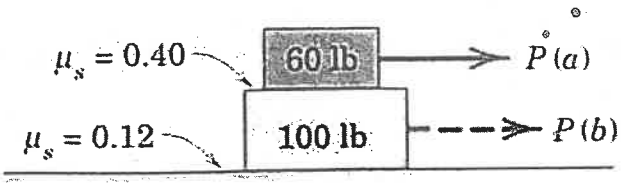
		subjected to a constant downward acceleration of 9.81 m/s^2 due to gravity. Neglect the effect of air resistance. Draw the diagram.			
6.	**a.**	The acceleration of a rocket traveling upward is given by a = (6 + 0.02s) m/s^2, where s is in meters. Determine the time needed for the rocket to reach an altitude of s = 100m. Initially, v = 0 and s = 0 when t = 0.	L4	CO5	**05**
		OR			
	b.	Write the equations of motion in the x and y directions for the 10-kg block. 	L3	CO5	**05**
7.	**a.**	The cable at A is being drawn toward the motor at V_A = 8m/s. Determine the velocity of the block 	L4	CO4	**05**
		OR			
	b.	At a given instant the 5-kg slender bar has the motion shown in Figure. Determine its angular momentum about point G and about the IC at this instant 	L4	CO5	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	The ratio of the lift force **L** to the drag force **D** for the simple airfoil is L/D = 10. If the lift force on a short section of the airfoil is 50lb, compute the magnitude of the resultant force R and the angle θ which it makes with the horizontal.	L4	CO1	10

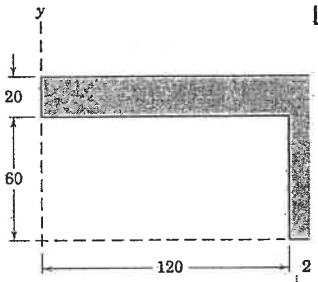
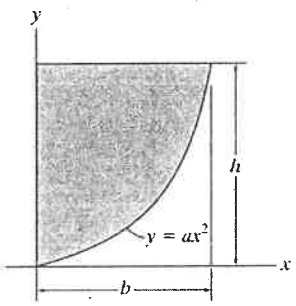
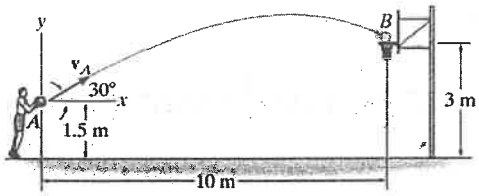


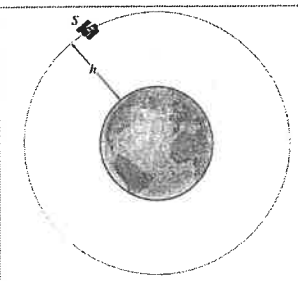
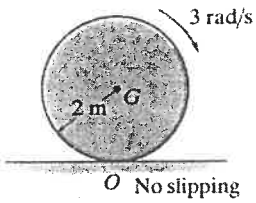
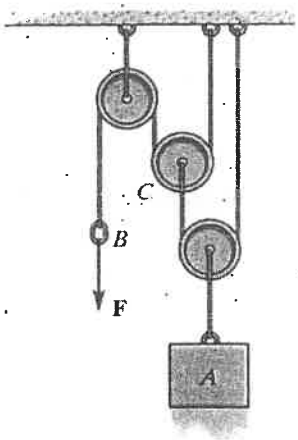
OR

	b.	The uniform 18kg bar OA is held in the position shown by the smooth pin at **O** and the cable AB. Determine the tension **T** in the cable and the magnitude and direction of the external pin reaction at O. 	L3	CO1	**10**
9.	**a.**	The horizontal position of the 500.kg rectangular block of concrete is adjusted by the 5° wedge under the action of the force **P**. If the coefficient of static friction for both pairs of wedge surfaces is 0.30 and if the coefficient of static friction between the block and the horizontal surface is 0.60, determine the least force **P** required to move the block. 	L4	CO2	**10**

OR

	b.	The force **P** is applied to (a) the 60-lb block and (b) the 100-lb block. For each case, determine the magnitude of **P** required to initiate motion. 	L3	CO2	**10**

10.	a.	Determine the product of inertia of the shaded area about the x-y axes. 	L3	CO3	10
		OR			
	b.	Determine the location the centroid x of the parabolic area. 	L3	CO3	10
11.	a.	The jet is traveling at a speed of 720 km/h. If the fuel is being spent at 0.8 kg/s, and the engine takes in air at 200 kg/s, whereas the exhaust gas (air and fuel) has a relative speed of 12000 m/s, determine the acceleration of the plane at this instant. The drag resistance of the air is $F_D = (55\ v^2)$, where the speed is measured in m/s. The jet has a mass of 7000kg.	L3	CO4	10
		OR			
	b.	The acceleration of a particle along a straight line is defined by $a = (2t - 9)\ m/s^2$, where **t** is in seconds. At t = 0, s = 1m and v = 10 m/s. When t = 9 s, determine (a) the particle's position, (b) the total distance traveled, and (c) the velocity.	L3	CO4	10
12.	a.	Determine the speed at which the basketball at A must be thrown at the angle of 30° so that it makes it to the basket at B. 	L4	CO5	10
		OR			

	b.	The satellite S travels around the earth in a circular path with a constant speed of 1000Km/h. If the acceleration is 2.5 m/s^2, determine the altitude h. Assume the earth's diameter to be 12,713 km.		L4	CO5	10
13.	**a.**	Determine the angular momentum of the 100-kg disk or rod about point G and about point O		L3	Co4	10
		OR				
	b.	If the supplied force F = 150 N, determine the velocity of the 50-kg block A when it has risen 3m, starting from rest. 		L4	CO5	10